[Metric]
AA-0-230E
December 31, 1997
Superseding
AA-0-230A/GEN
October 7, 1964

FEDERAL SPECIFICATION

OFFICE FURNITURE: COSTUMER, EXECUTIVE TYPE, WOOD, TRADITIONAL

This Federal Specification was developed by the National Furniture Center, Federal Supply Service, General Services Administration, Washington, DC 20406 based upon currently available technical information.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the general and detailed requirements for Executive type, traditional office wood costumers.

2 MATERIALS

2.1 Adhesives. The adhesives used in the costumer shall be any that will develop an average block shear strength of 19 300 kPa minimum.(4.3.1).

2.2 Finish. The finish shall match the standard laminate samples in color with a satin finish and meet the test in 4.3.2. It shall be uniform, smooth, with no discoloration, grit, tackiness, peeling, flaking, blistering, cracking, damage or rub through.

2.4 Wood. The following wood species are permitted:

Cherry
Mahogany 1/
Walnut
Andiroba (Carapa guianensis)

1/ Khaya/African or Swietenia/American.

2.4.1 Solid wood requirements. Wood used for exposed parts, shall be bright, well sanded, and free from brashness, discoloration, worm holes, splits, shake, or other defects which affect the appearance or which may affect the serviceability of the finished product.

2.4.2 Wood seasoning. All wood, shall be uniformly kiln dried to a 5 to 8 percent moisture content as determined by the oven-drying method. After drying, it shall be held in temperature and humidity controlled storage for a sufficient time to ensure uniform moisture distribution among the individual pieces.

2.5 Hardware.

2.5.1 Face Hardware. The coat hooks, shall be made of cast brass. The castings shall be true, without air or sand holes or other defects. All ornamentation on face of the hardware, such as mouldings, beadings, or turnings, shall have a clean and even surface. All surfaces shall be smoothly and evenly finished. Hardware shall receive minimum one coat of baked synthetic clear finish. (See Figure 5)

2.5.2 Nail Glides. Glides shall be 22 mm minimum diameter, rubber cushioned nail type. The base shall be brown high impact plastic.

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2.5.3 Other hardware. All screws, whose heads are exposed on brass hardware shall be brass.

3. CONSTRUCTION

3.1 Design and construction. The design and construction of the costumers shall be as specified herein and as shown in the Figures. In any case where there is a conflict between the text of the specification and the Figures, the text shall prevail.

3.2 Tolerances. Where tolerances are not specified, the following tolerances are permitted:

Overall dimensions: plus, minus 6.5 mm.
Solid wood part dimensions: plus 4 mm, minus 2 mm.

3.3 Shaft. The shaft shall be made up of two parts, joined at the middle with a minimum 7 mm x 89 mm hanger bolt and minimum 19 mm long internal and external threaded insert as shown in figure 2. Both parts shall be made either of solid or built-up stock which shall be carefully matched with other parts of the costumer for color and wood grain pattern. The shaft shall be reeded as shown in figure 1 and 5. Four hat and coat hooks (see 2.5.1 and figure 5) shall be attached to the top of the shaft with two minimum 22 mm long brass screws and positioned as shown in figure 1.

3.4 Base. The base shall consist of two crossed supports, four filler pieces, four braces, and four feet. The crossed supports shall be 540 mm x 63 mm W x 32 mm thick, and lap-joined at the center. The filler pieces shall be securely glued to the cross supports at their intersection. The assembly shall be shaped and molded as shown in figures 1, 2, 3, 4 and 5. The braces shall be shaped and molded as shown in figures 3 and 4. The braces shall be glued and doweled or tenoned to the shaft, and glued and screwed to the cross supports. The feet shall be turned as shown in figure 4 and shall be glued and doweled to the cross- supports. A minimum 9 mm x 76 mm lag screw, with flat washer shall be used to secure the base to the shaft.

3.5 Joinery. All joints shall be true, securely glued (3.2.1). Starved or insufficiently glued joints will not be acceptable.

3.6 Dowels. Dowels shall be made of maple, birch, hickory, or beech, dried to a moisture content of 3 to 5 percent. They shall have longitudinal or spiral grooves and shall be beveled on each end.

3.7 Built-up stock. Built-up stock used for exposed components in the construction of the costumer shall consist of not more than three full length pieces, each of which shall be approximately the same thickness. They shall be carefully matched for color and figure.

3.8 Finish, exposed wood parts and surfaces. All surfaces exposed to view in normal use shall be finished, sanded smooth with all corners and edges eased. Exposed wood finish shall match Federal Walnut standard sample FSS-L-01024 or Federal Mahogany FSS-L-01025 (6.4) with a satin finish. The natural grain of the wood shall not be clouded by the finishing materials. Bleaching agents or materials shall not be used. The application of materials, drying time, sanding, cleaning, and rubbing shall be controlled to produce items of uniform finish without sags, runs, orange peel or other defects detrimental to a smooth, quality appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection Provisions. Details of the first article sample requirements are contained in the solicitation. In addition, approval of the first article sample authorizes the start of production, but does not relieve the contractor of responsibility from conformance with all other applicable provisions of this specification. Inspection requirements are stated in the solicitation.

4.2 Testing.

Component	Characteristic	Reference	Test Method
Adhesive	Block shear test	2.1	ASTM D 905-94
Finish	Toughness and adhesion	2.2	4.3.2 herein

4.3 Test methods.

4.3.1 Test for Adhesives. Test in accordance with ASTM D 905. Rerun test if the average shear strength of all samples is below 19 300 kPa; there is a 10 percent or greater difference between high and low specimen values, and at least one test specimen broke at more than 19 300 kPa.

Disregard a test specimen in computing the average if it breaks at less than 19 300 kPa; and it has 50 percent or more wood failure.

4.3.2 Test for Finish. The toughness and adhesion test shall be made by using Organic Coating Adhesion Tester, US Testing Company, Model Number 1001, in accordance with instructions for using, as provided for by the manufacturer of the tester (see 6.5). The panel shall be marred both parallel and perpendicular to the grain. The resulting indentation shall be examined for whitening (film separation)or cracking which is not acceptable.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing, Marking. Package, pack, and mark shipping containers in accordance with the contract or order.

6. NOTES

6.1 Intended use. The traditional executive costumers are designed for use in executive offices, courtrooms, libraries, and conference rooms of Federal agencies.

6.2 Regulatory Requirements. The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580, as amended, to the maximum extent practicable.

6.3 Ordering data. Purchasers should include the title, number, and date of this specification, quality assurance provisions, and packaging, packing and marking requirements in procurement documents:

6.4 Standard plastic laminate sample. Standard plastic laminate samples (Federal Walnut FSS-L-01024 and Federal Mahogany FSS-L-01025) may be obtained without charge from:

National Furniture Center
Engineering Division
GSA/FSS (3FNE)
Washington, DC 20406

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6.5 Organic Coating Hardness Tester. Information about the Organic Coating Hardness Tester, Model NO. 1001, may be obtained from:

(US Testing Company, Inc. Instrument Marketing Division 1415 Park Avenue Hoboken, NJ 07030)

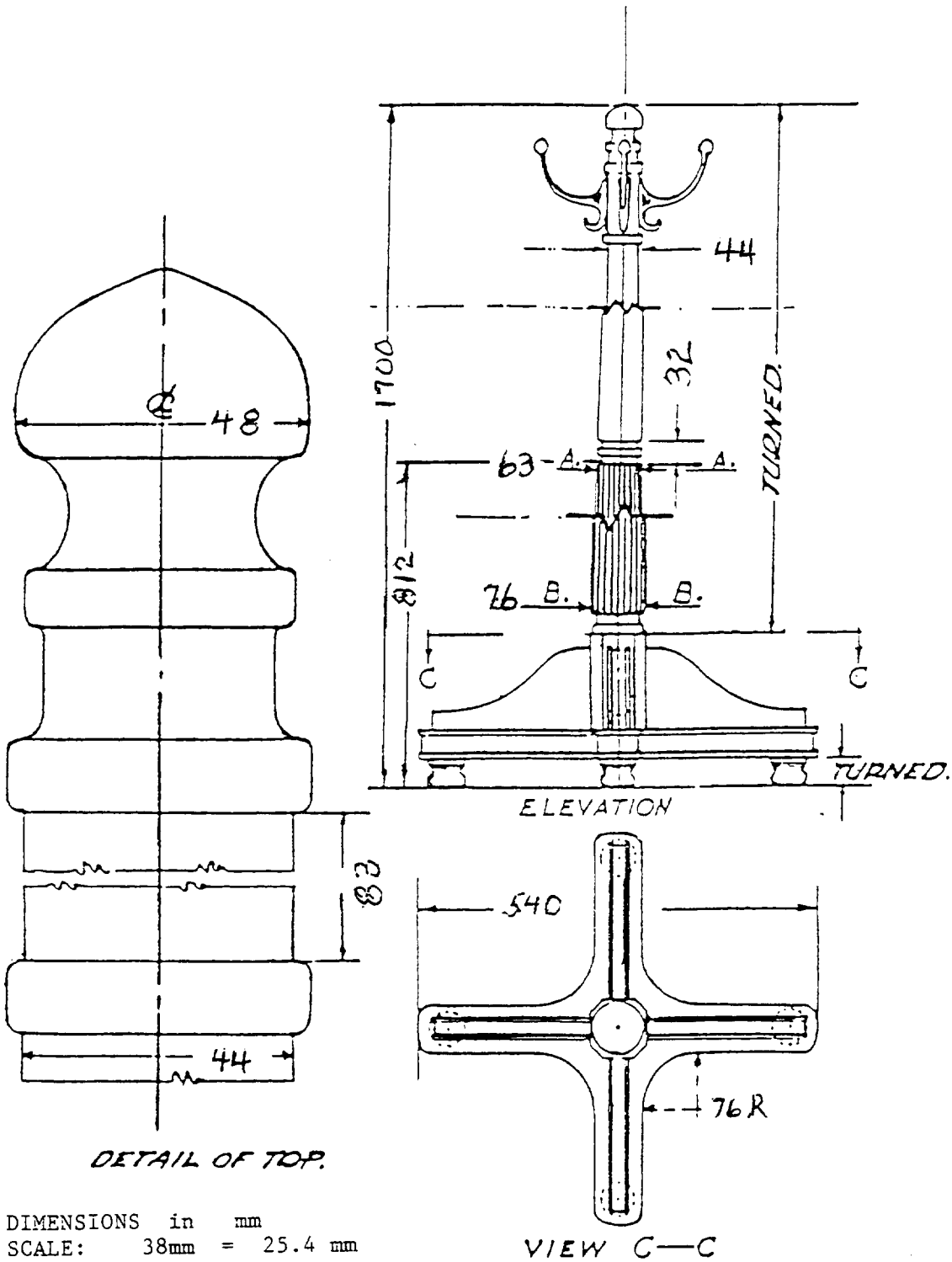
6.6 American Society for Testing and Materials (ASTM) Standard:

ASTM D 905-94 - Standard Method of Test for Strength Properties of Adhesives in Shear by Compression Loading.

(Copies may be obtained from American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2925.)

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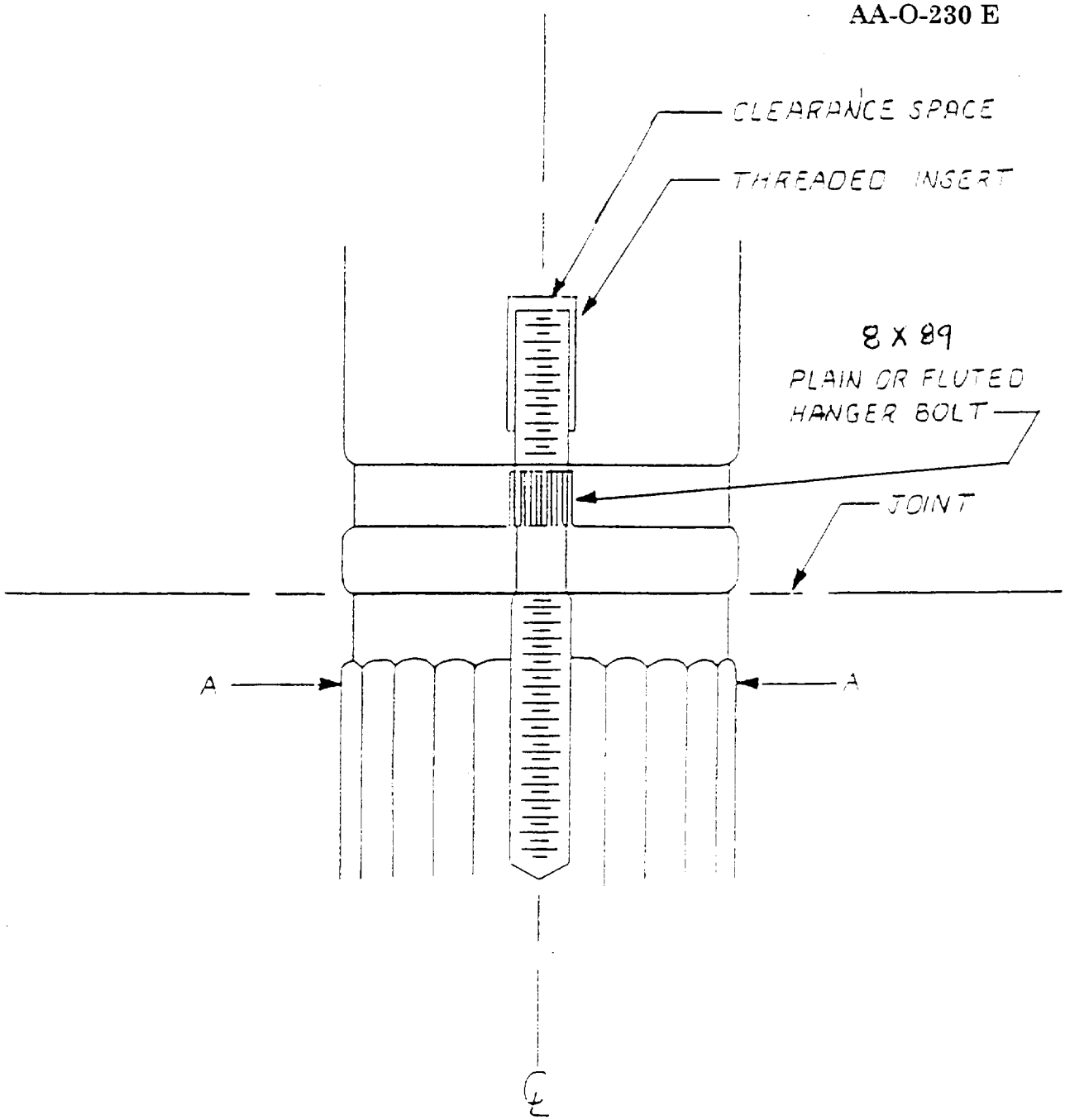
FIGURE 1. COSTUMER



DIMENSIONS in mm
SCALE: 38mm = 25.4 mm

Overall height without glides.

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DIMENSIONS in mm

FIGURE 2. Center Joint Section of Shaft (NOT TO SCALE)

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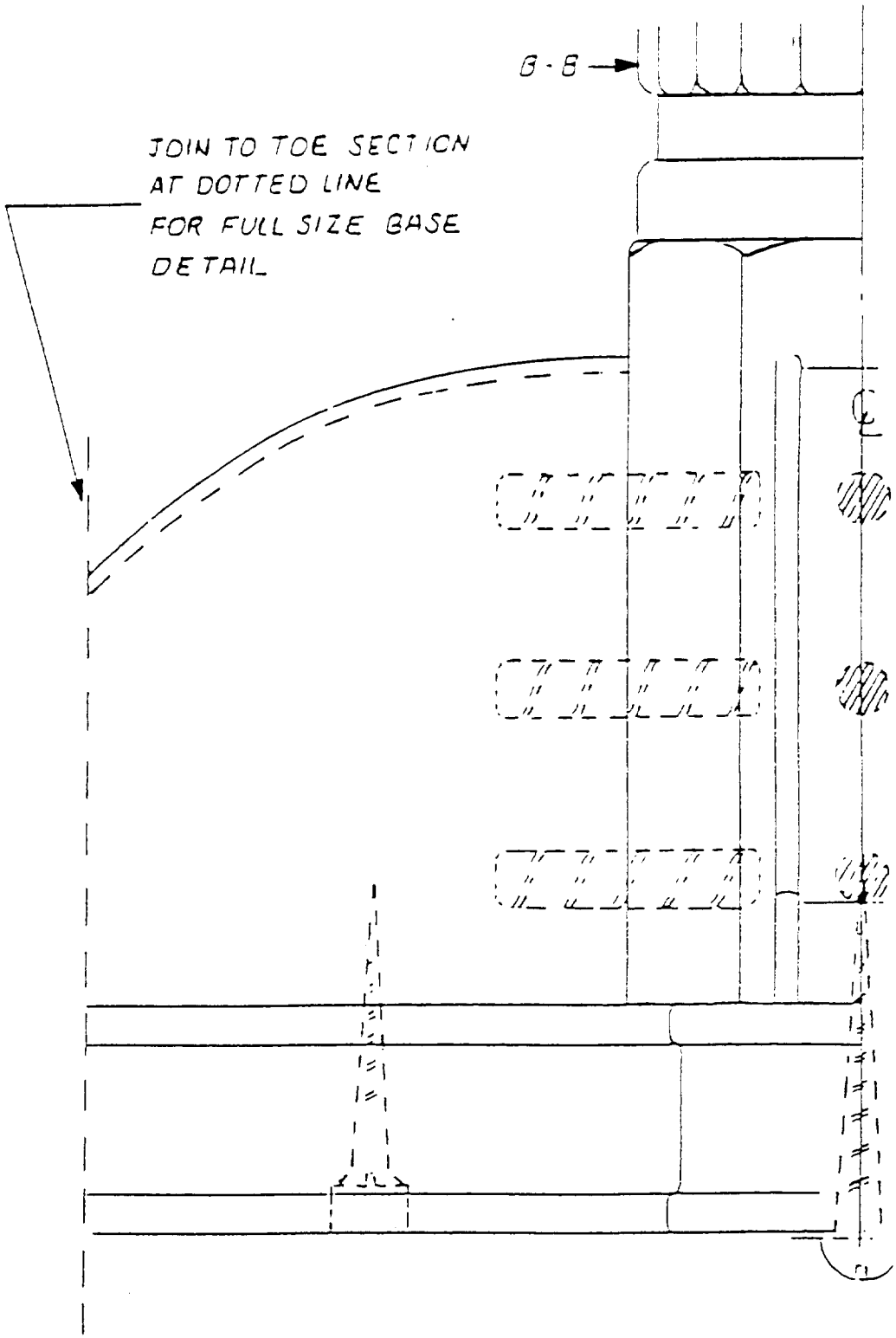
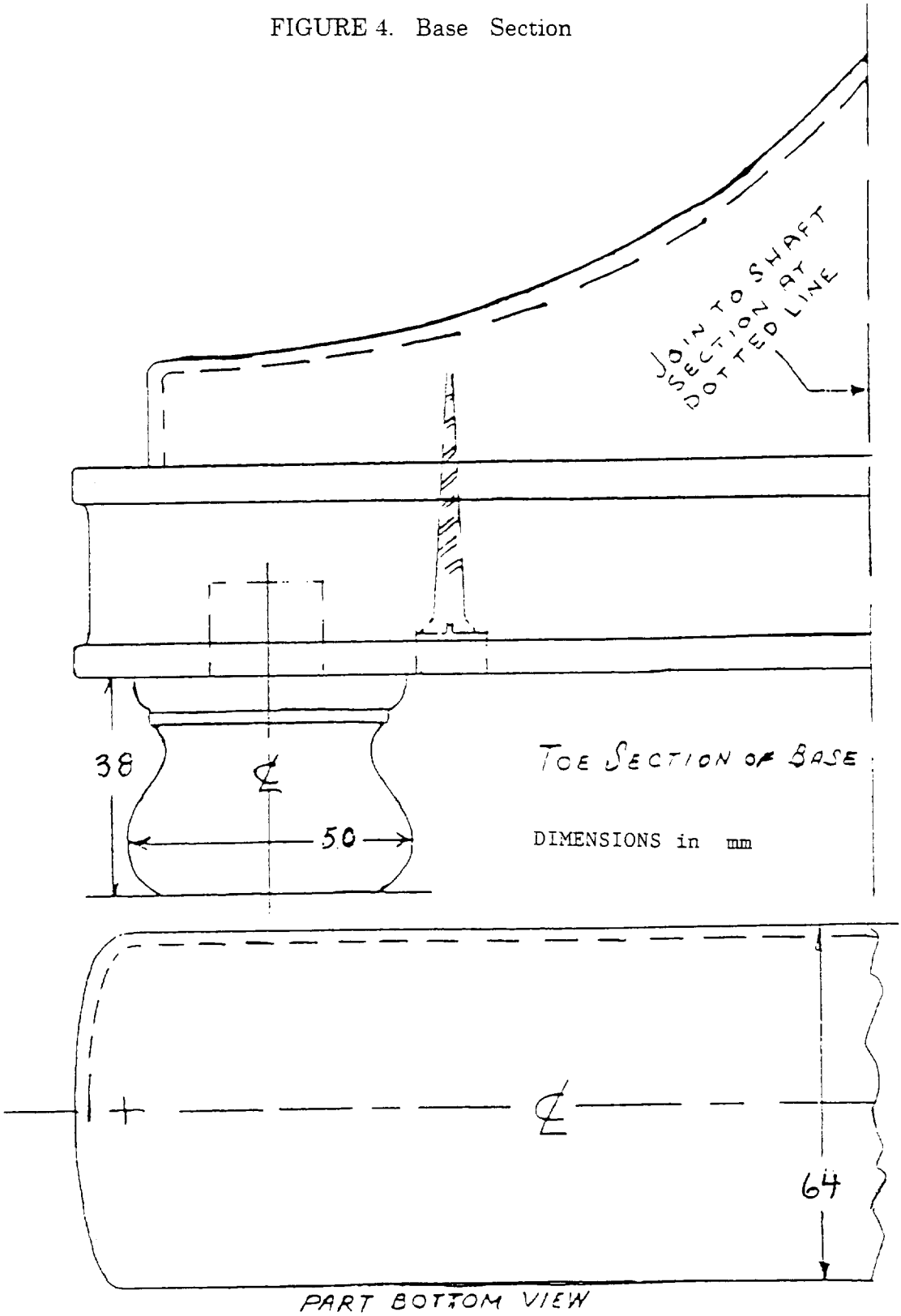


FIGURE 3. Base Section

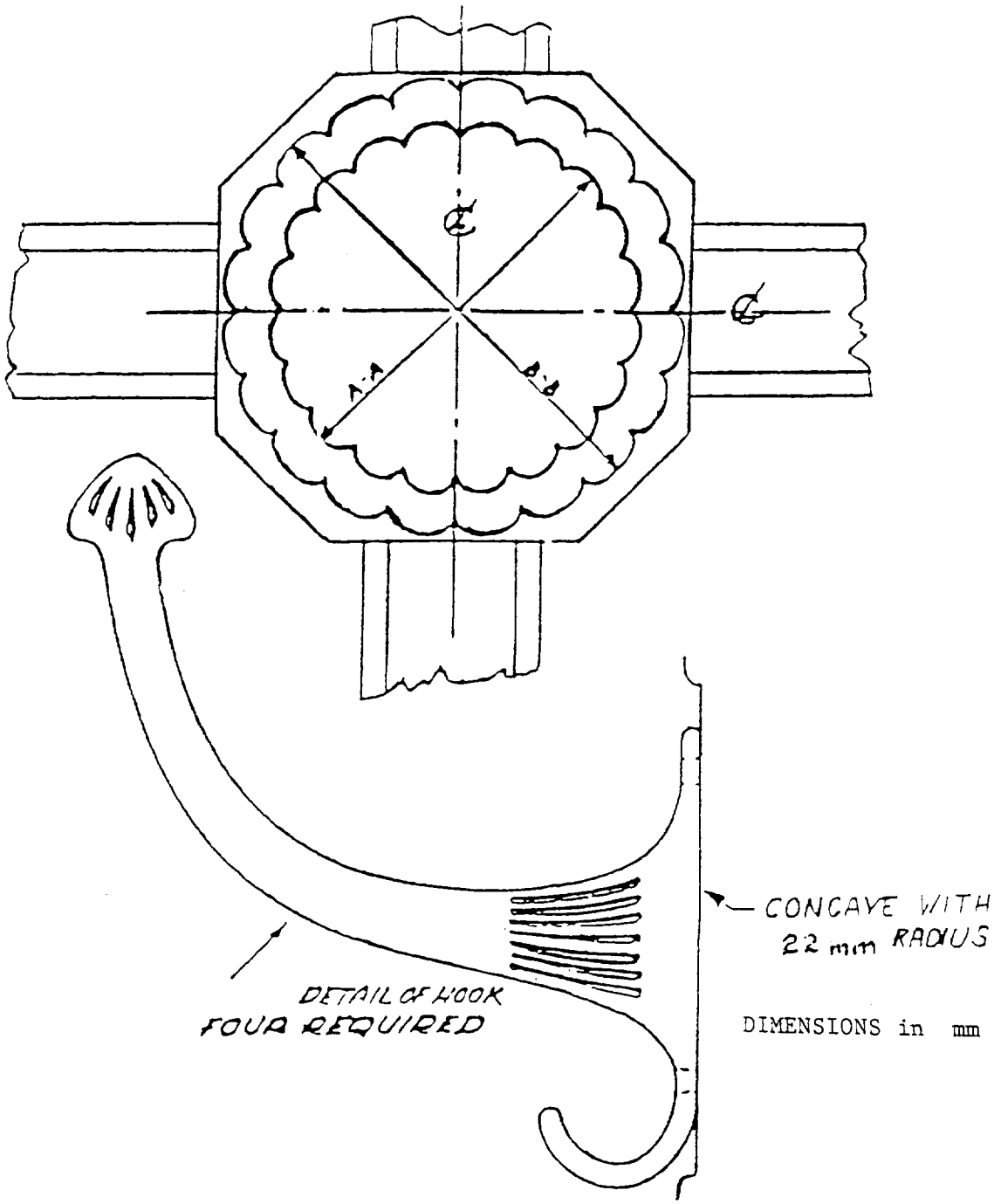
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FIGURE 4. Base Section



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FIGURE 5. Shaft Cross Section and Hook



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